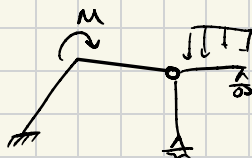


TEMA 3. ESTRUCTURAS RETICULADAS

→ Con que 1 sólo nodo sea rígido, ya es reticulada.

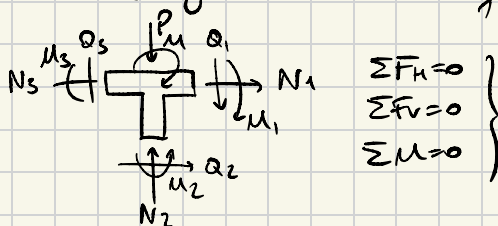
1.- Nodos rígidos



2.- Momentos muy pequeños.

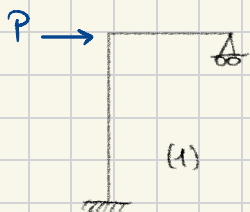
3.- Infinitamente rígidos a axiales.

4.- Todos los nodos rígidos están en equilibrio.

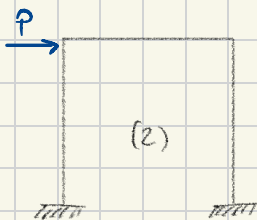


5.- Clasificación

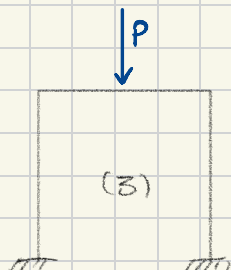
- A) Translacional: Si una sola barra se traslada, ya es translacional. Deben moverse los dos extremos. Se resuelven como en el T.1.
- B) Intraslacional



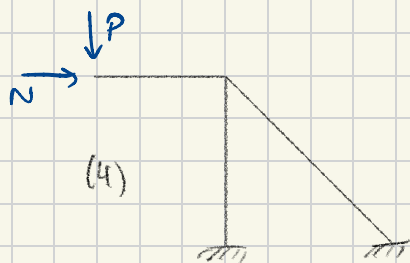
TRASLACIONAL



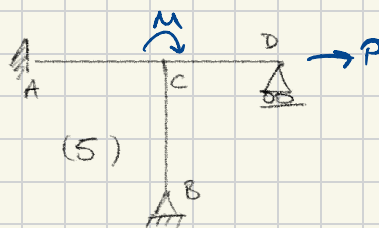
TRASLACIONAL



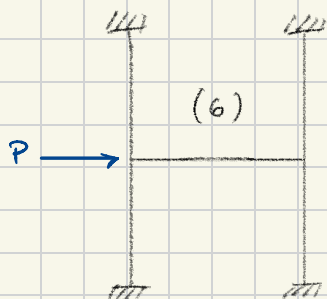
INTRASLACIONAL



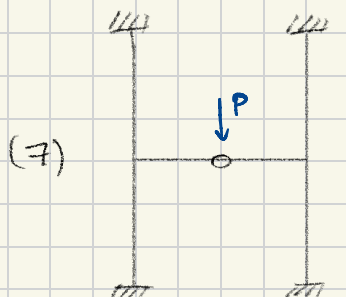
INTRASLACIONAL



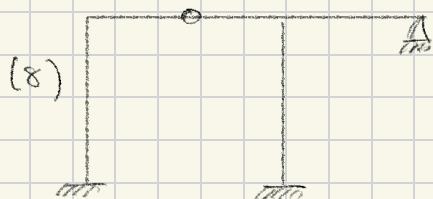
INTRASLACIONAL



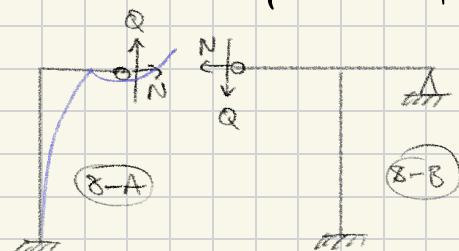
TRASLACIONAL



INTRASLACIONAL



Si hay rótula, partimos en dos partes:



TRASL.

INTRASL.

Se resuelve cada una por separado.

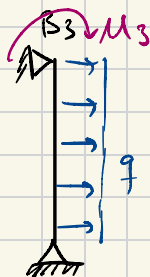
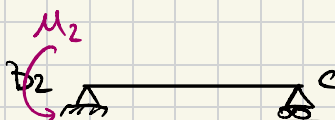
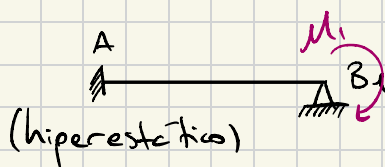
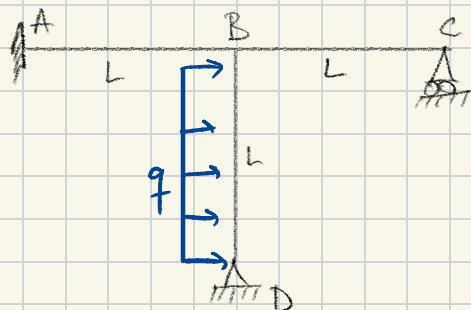
ESTRUCTURAS INTRASLACIONALES

RESOLUCIÓN

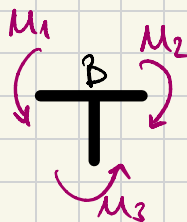
a) Clarificación:
→ Intraslacional.

b) Resolución.

Como ninguna barra se desplaza, las aíslas.



Se ponen las reacciones al contrario de cómo los pusimos



$$\sum M = 0 \quad [1] \quad M_3 + M_2 + M_1 = 0$$

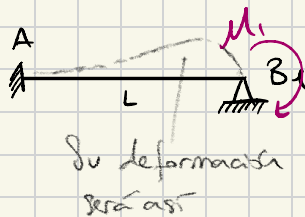
$$\vec{\theta}_{B1} = \vec{\theta}_{B2} \quad [2]$$

$$\vec{\theta}_{B1} = \vec{\theta}_{B3} \quad [3]$$

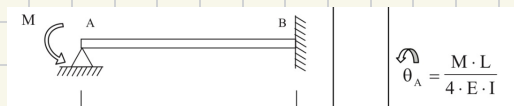
$$\vec{\theta}_{B2} = \vec{\theta}_{B3}$$

Cualquiera de las tres relaciones va a dar lo mismo. Es el mismo nodo.

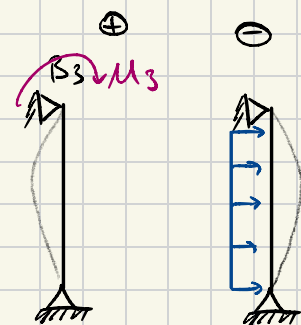
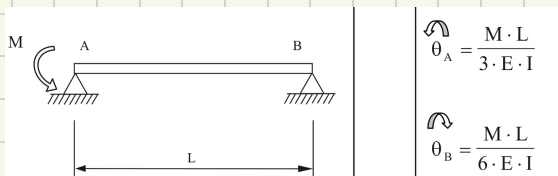
Los gnos los calculamos con el formulario



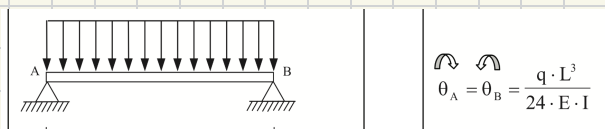
$$\theta_{B1} = + \frac{M_1 \cdot L}{4EI}$$



$$\theta_{B2} = - \frac{M_2 \cdot L}{3 \cdot EI}$$



$$\theta_3 = + \frac{M_3 \cdot L}{3EI} - \frac{qL^3}{24EI}$$



$$[1] \quad M_3 + M_2 + M_1 = 0$$

$$\theta_{B1} = \theta_{B2} [2] \Rightarrow \frac{M_1 L}{4EI} = - \frac{M_2 L}{3EI} \rightarrow 3M_1 + 4M_2 = 0 [2]$$

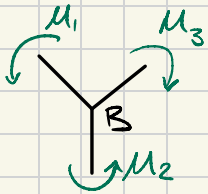
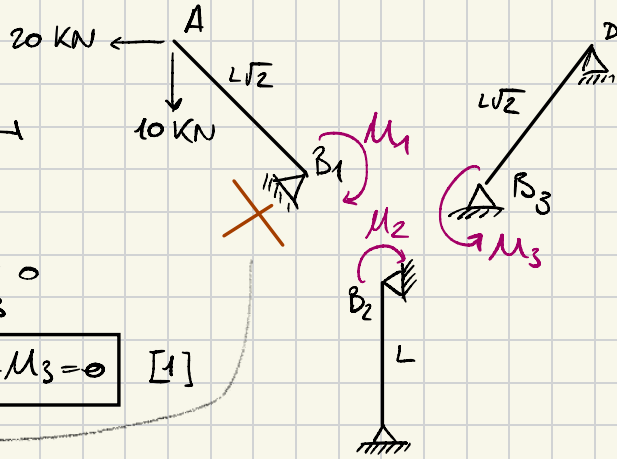
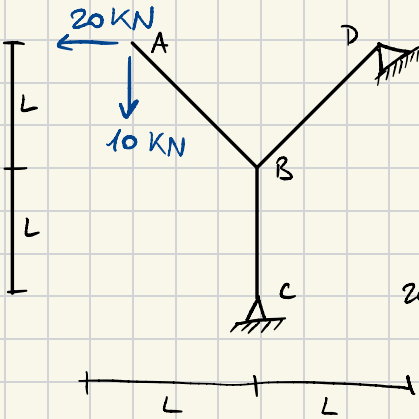
$$\theta_{B1} = \theta_{B3} [3] \Rightarrow \frac{M_1 L}{4EI} = \frac{M_3 L}{3EI} - \frac{qL^3}{24EI} \rightarrow 6M_1 - 8M_3 = -qL^2 [3]$$

⊗ PROBLEMA

a) ¿Trasl./Intras.?

INTRASLACIONAL

- b) Mov. en B.
c) Ley Flectores
d) Reacciones



$$\sum M_{\text{nudo } B}^{\uparrow} = 0$$

$$M_1 + M_2 - M_3 = 0 \quad [1]$$

$$\sum M_{B_1}^{\uparrow} = 0 \rightarrow M_1 - 20L - 10L = 0$$

$$M_1 = 30L$$

Iguando giros: $\theta_{B_2} = \theta_{B_3}$

Formulario

$$\frac{M_2 \cdot L}{3EI} = -\frac{M_3 \cdot L\sqrt{2}L}{3EI} \Rightarrow M_2 + \sqrt{2}M_3 = 0 \quad [2]$$

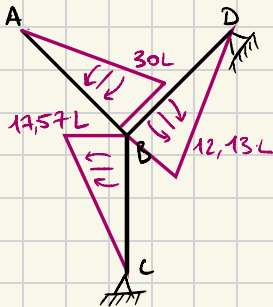
Resolviendo [1] y [2]:

$$\begin{aligned} M_1 &= 30L \\ M_2 &= -17,57L \\ M_3 &= 12,43L \end{aligned}$$

b) Por intranslacional $\vec{u}_B = 0, \vec{v}_B = 0$.

$$\vec{\theta}_B = \vec{\theta}_{B_2} = \frac{M_2 L}{3EI} = \frac{-17,57 L^2}{3EI}$$

c)



d)

Trans B₂-C

$$\sum \vec{M}_{B_2}^+ = 0 \rightarrow H_C \cdot L - M_2 = 0$$

$$H_C = -17,57$$

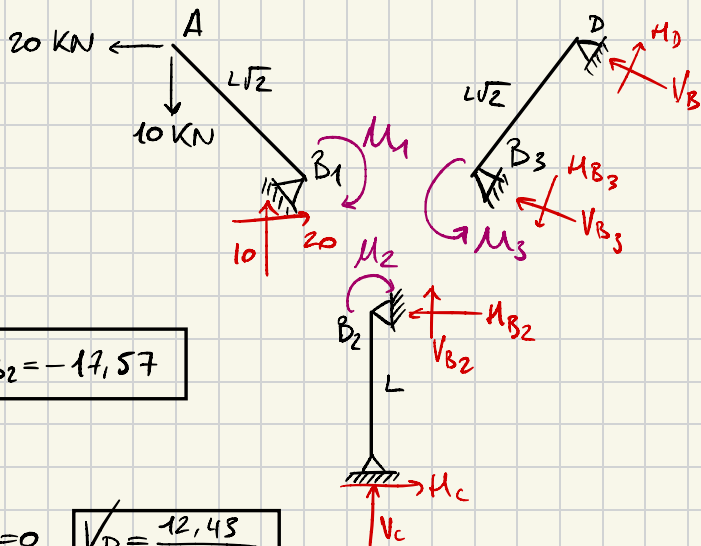
$$\sum F_H = 0 \quad H_C - H_{B_2} = 0$$

$$H_{B_2} = -17,57$$

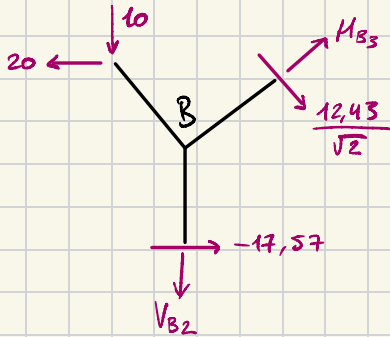
Trans B₃-C

$$\sum \vec{M}_{B_3}^+ = 0 \quad M_3 - \sqrt{2} \cdot L \cdot V_D = 0 \quad V_D = \frac{12,43}{\sqrt{2}}$$

$$\sum F_V^+ = 0 \quad V_3 - V_D = 0 \quad V_3 = \frac{12,43}{\sqrt{2}}$$



Equilibrio del nodo



$$\sum F_H = 0 \quad -20 + H_{B2} + \frac{V_{B3}}{\sqrt{2}} + \frac{H_{B3}}{\sqrt{2}} = 0$$

$$\sum F_V = 0 \quad 10 + V_{B2} + \frac{V_{B3}}{\sqrt{2}} - \frac{H_{B3}}{\sqrt{2}} = 0$$

$$H_{B3} =$$

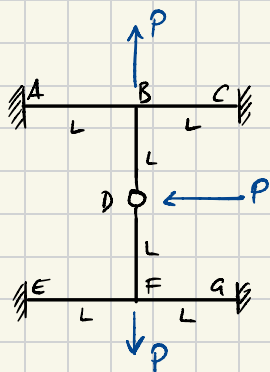
$$V_{B2} =$$

Calcular V_c y H_D

Tramo B₂-C $\sum F_V^{\uparrow} = 0 \quad V_c + V_{B2} = 0$

Tramo B₃-D $\sum F_V^{\uparrow} = 0 \quad H_D - H_{B3} = 0$

* PROBLEMA 3.21

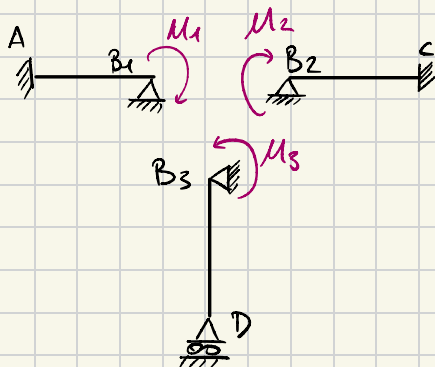
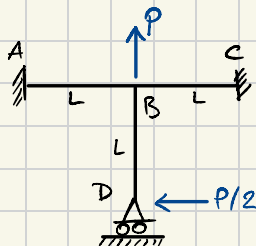


a) ¿Trasl. / luras?

b) Ley de cortantes y flectores AB, BC, BD.

c) M_D ; V_D ; θ_{rel} .

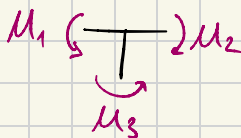
a) luraslacional. Simétrica.



$$\sum M_{B3}^+ = 0 \quad M_3 - \frac{P}{2}L = 0$$

$$\boxed{M_3 = \frac{PL}{2}}$$

Nudo:



$$\sum M_{nudo}^+ = 0$$

$$M_1 + M_2 - M_3 = 0$$

$$\boxed{M_1 + M_2 = \frac{PL}{2}} \quad [1]$$

Giros en nudo B

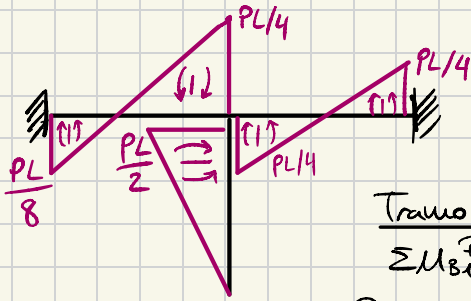
$$\theta_{B1} = \theta_{B2}$$

$$\frac{M_1 \cdot L}{4EI} = \frac{M_2 \cdot L}{4EI}$$

$$\boxed{M_1 = M_2} \quad [2]$$

$$\boxed{M_1 = M_2 = \frac{PL}{4}}$$

b) Flectores



Tramo B2-C

$$\sum M_{B2}^+ = 0$$

$$M_2 + \frac{M_2}{2} - R_c \cdot L = 0 \rightarrow$$

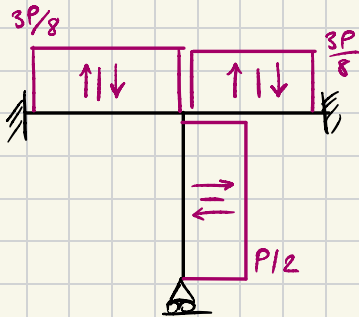
Tramo A-B1

$$\sum M_{B1}^+ = 0$$

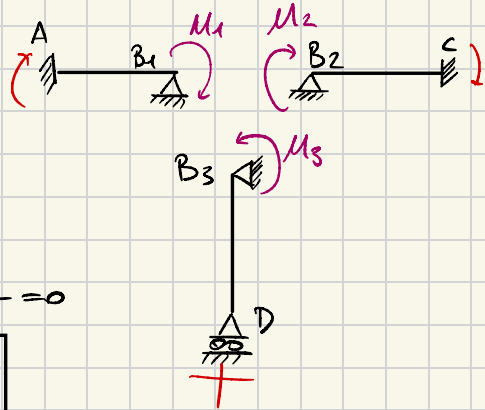
$$R_a L + M_1 + \frac{M_1}{2} = 0$$

$$R_a = -\frac{3P}{8}$$

$$R_c = \frac{3P}{8}$$



Constantes



c) M_D ; V_D ; θ_{rot}^{rel}

$$\uparrow V_D = 0$$

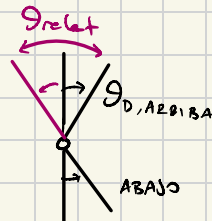
$$\overline{u}_D = \overline{u}_B + \overline{\theta}_B \cdot L + \frac{1}{EI} \left[\frac{L}{2} \cdot \frac{PL}{2} \cdot \frac{2}{3} L \right]$$

$$\overline{\theta}_B = \frac{M_1 \cdot L}{4EI} = \frac{PL^2}{16EI}$$

$$\overline{u}_D = \frac{PL^3}{EI} \left(\frac{1}{6} + \frac{1}{16} \right)$$

$$\overline{\theta}_{D, AB21BA} = \overline{\theta}_B + \frac{1}{EI} \left[\frac{L}{2} \cdot \frac{PL}{2} \right] =$$

$$= \frac{PL^2}{16EI} + \frac{PL^2}{4EI} = \frac{5PL^2}{16EI}$$



$$\theta_{rel}^{rel} = \frac{5PL^2}{8EI}$$

Al ser simétrica,
solo hace falta
hacer un lado.

⊕ PROBLEMA 3.1

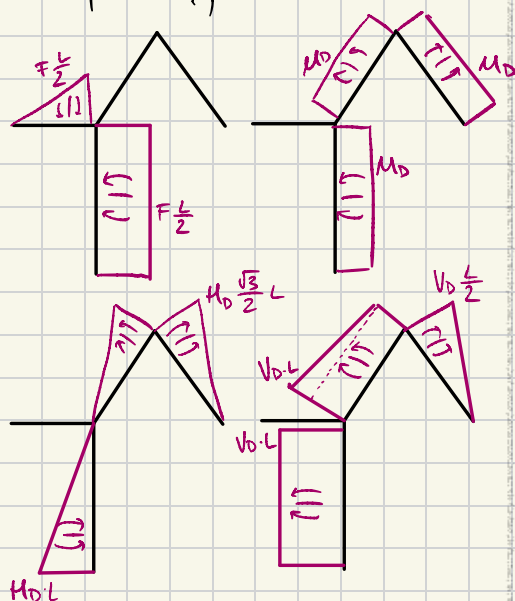
$$L = 2\text{ m}$$

$$F = 10\text{ kN}$$

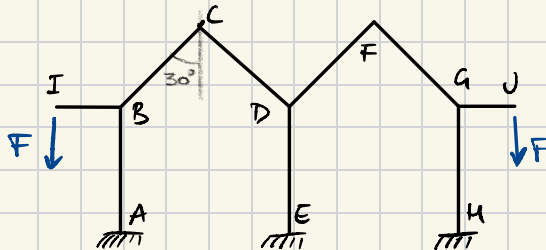
$$EI = 5 \cdot 10^6\text{ kNm}^2$$

- Esfuerzos en $\overline{DE}$?
- Mov. D
- Reacciones
- Mov. B
- Despl. relat. I-J

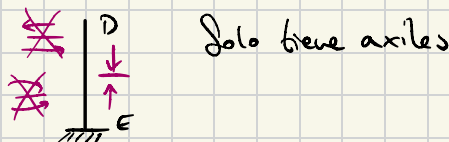
Ley de flectores



$$\begin{aligned} \overline{\theta}_D = 0 &= \dots \Rightarrow 6M_D + 1.464H_D + 8V_D = -20 \\ &[1] \\ \uparrow \overline{V}_D = 0 &= \dots \Rightarrow 8M_D - 0.536H_D + \frac{40}{3}V_D = -40 \\ &[2] \\ \rightarrow \overline{U}_D = 0 &= \dots \Rightarrow 1.464H_D + \frac{20}{3}V_D - 0.536V_D = 20 \\ &[3] \end{aligned}$$



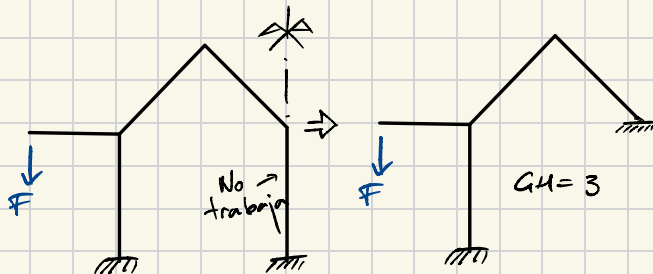
a) y b) Barra D-E es simétrica.



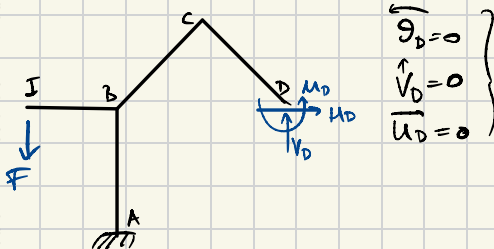
$$\begin{aligned} \overline{U}_D = 0 \\ \overline{\theta}_D = 0 \\ \overline{V}_D = 0 \end{aligned} \left. \begin{array}{l} \text{por E al eje simetr.} \\ \text{por D rígida.} \end{array} \right\}$$

c) Reacciones

Traslacional. $GH = 6$



Resolución

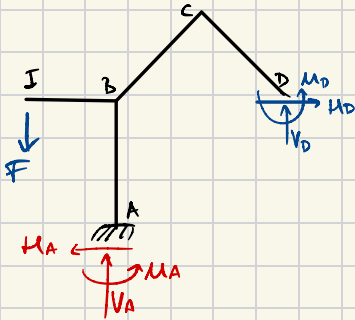


$$\begin{aligned} \overline{\theta}_D = 0 \\ \overline{V}_D = 0 \\ \overline{U}_D = 0 \end{aligned} \left. \begin{array}{l} \text{por E al eje simetr.} \\ \text{por D rígida.} \end{array} \right\}$$

→ S.E. [1][2][3] GABLO:

$$\begin{aligned} M_D &= -1,31 \text{ kNm} \\ H_D &= 3,12 \text{ kN} \\ V_D &= -2,09 \text{ kN} \end{aligned}$$

Reacciones:

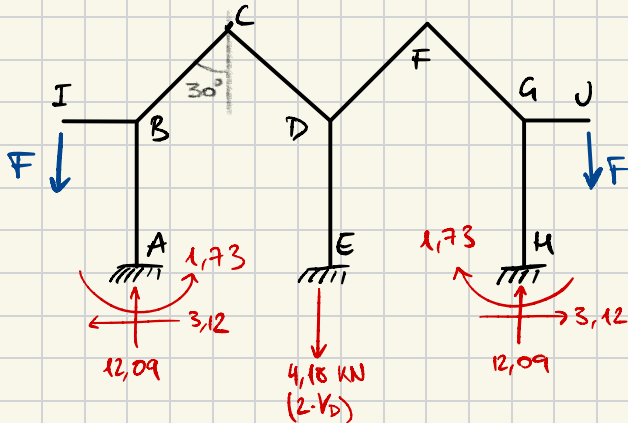


$$\sum M_A^+ = 0 \rightarrow M_A + F \cdot \frac{L}{2} + M_D + V_D L - H_D L = 0$$

$$M_A = 1,73 \text{ kNm}$$

$$\sum F_H = 0 \quad H_D - H_A = 0 \rightarrow H_A = H_D = 3,14 \text{ kN}$$

$$\sum F_V = 0 \quad V_A + V_D - F = 0 \rightarrow V_A = 12,09 \text{ kN}$$



d) Mov. en B

$$V_B = 0$$

$$\theta_B = \frac{1}{EI} \left[L \cdot F \cdot \frac{L}{2} + L \cdot M_D - \frac{L}{2} H_D L + L \cdot V_D \cdot L \right]$$

$$\theta_B = 5,56 \cdot 10^{-4} \text{ rad}$$

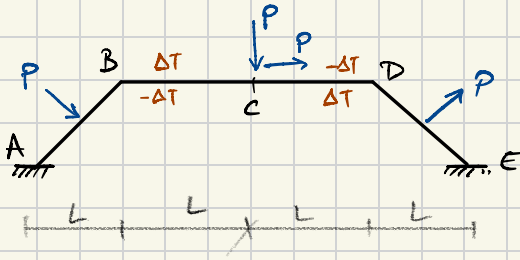
$$\begin{aligned} \Delta_B = \frac{1}{EI} \left[L \cdot F \cdot \frac{L}{2} \cdot \frac{L}{2} + M_D \cdot \frac{L}{2} - \frac{L}{2} H_D L \cdot \frac{2}{3} L + \right. \\ \left. + L \cdot V_D L \cdot \frac{L}{2} \right] \end{aligned}$$

$$\Delta_B = 1,4 \cdot 10^{-4} \text{ m}$$

e) Despl. rel + I-J

$$\delta_{I,J} = 2\Delta_B = 2,8 \cdot 10^{-4} \text{ m}$$

PROBLEMA 3.13

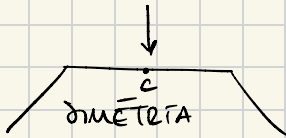


$$C = L/10$$

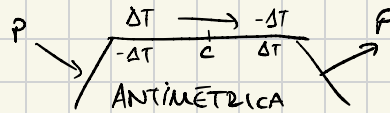
$$\alpha = 10^{-5}/\Delta T$$

$$EI = 100 PL^2$$

Lo separamos en dos esquemas:



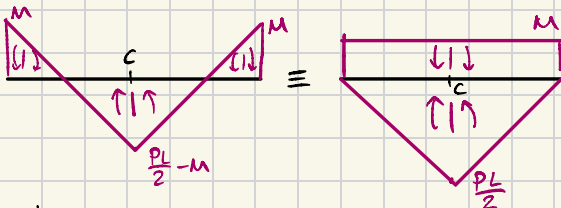
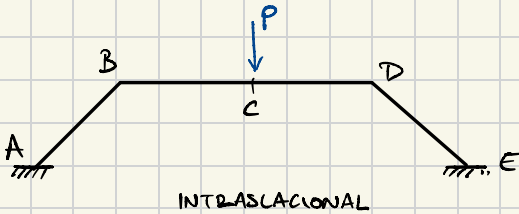
De aquí sacamos V_c



De aquí sacamos U_c, θ_c

SUPERPOSICIÓN

b) Mov. C



$$V_c = \theta_B \cdot L + \frac{1}{EI} \left[L \cdot M \cdot \frac{L}{2} - \frac{L}{2} \cdot \frac{PL}{2} \cdot \frac{L}{3} \right] = 78,84 \cdot 10^{-5} \cdot L \text{ [m]}$$

La mitad del área (C en la mitad)

$$V_c = 78,84 \cdot 10^{-5} \cdot L \text{ [m]}$$

a) GH; TRABL; SIMETR.

b) Mov. C

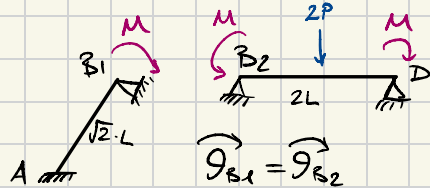
c) Reacciones

d) Diagramas

a) GH = 6 - 3 = 3

Traslacional (B-D)

No simetría ni antisimetría.



$$\theta_{B1} = \theta_{B2}$$

$$\frac{M\sqrt{2}L}{4EI} = -\frac{M \cdot 2L}{3EI} - \frac{M \cdot 2L}{6EI} + \frac{P \cdot L \cdot L(2L+L)}{6EI \cdot 2L}$$

$$M = \frac{PL}{\sqrt{2} + 4} = 0,19 \cdot PL$$

→ Calculamos sus reacciones:

$$\sum M_A = 0 \quad 2\sqrt{2}L - \frac{M}{2} - M = 0 \rightarrow R = 0,196P$$

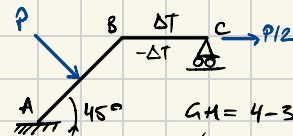
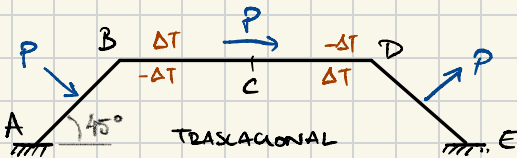
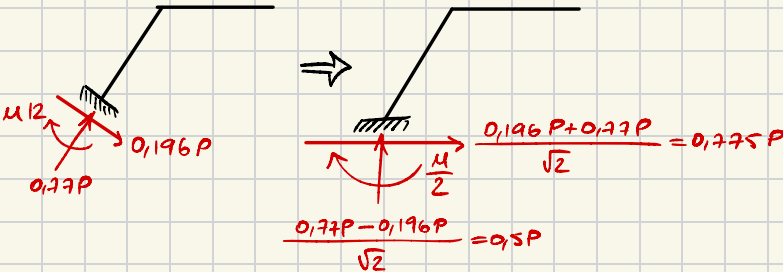
Ej. nodo B:

$$\sum F_v = 0 \quad \frac{N_1}{\sqrt{2}} - \frac{R}{\sqrt{2}} - \frac{P}{2} = 0$$

$$N_1 = 0,9P$$

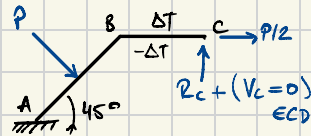
$$\sum F_H = 0 \quad \frac{N_1}{\sqrt{2}} + \frac{R}{\sqrt{2}} - N_2 = 0$$

$$N_2 = 0,77P$$



$$GH = 4 - 3 = 1.$$

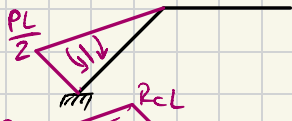
Llevamos al canto



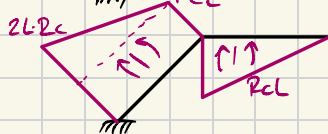
Ley de fletores



Por P ↓



Por $\frac{P}{2} \rightarrow$



Por $2c \uparrow$

$$\begin{aligned} \sum V_c = 0 = & \delta_A + \frac{1}{EI} \left[-\frac{1}{2} \frac{L\sqrt{2}}{2} P \frac{\sqrt{2}}{2} \left(\frac{3L}{2} + \frac{2}{3} \frac{L}{2} \right) - \right. \\ & - \frac{1}{2} L\sqrt{2} \cdot \frac{P}{2} \left(L + \frac{2}{3} L \right) + L\sqrt{2} \cdot R_c L \cdot \frac{3L}{2} + \\ & + \frac{1}{2} \sqrt{2} \cdot L \cdot R_c L \left(L + \frac{2}{3} L \right) + \frac{1}{2} L \cdot R_c \cdot L \cdot \frac{2L}{3} \left. \right] - \\ & - 2 \cdot \frac{2\Delta T}{c} \cdot L \cdot \frac{L}{2} = 0. \rightarrow R_c = 0,564P \end{aligned}$$

$$\vec{M}_C = \frac{1}{EI} \left[+\frac{1}{2} \frac{\sqrt{2}}{2} PL \frac{\sqrt{2}}{2} \left(\frac{2}{3} \frac{L}{2} + \frac{L}{2} \right) + \frac{1}{2} \sqrt{2} \cdot L \frac{PL}{2} \left(\frac{2L}{3} \right) - \sqrt{2} R_C \cdot L \frac{L}{2} - \frac{1}{2} \sqrt{2} L R_C L \frac{2}{3} L \right]$$

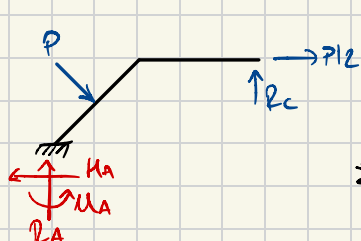
$$\rightarrow \boxed{\vec{M}_C = -2,2 \cdot 10^{-3} \cdot L}$$

Possivelmente está mal.

No tiene sentido.

$$\vec{\theta}_C = \frac{1}{EI} \left[-\frac{1}{2} \frac{\sqrt{2}}{2} PL \frac{\sqrt{2}}{2} - \frac{1}{2} \sqrt{2} L \frac{PL}{2} + \sqrt{2} R_C L + \frac{1}{2} \sqrt{2} L R_C L \right] -$$

$$-\alpha \cdot \frac{2\Delta T}{C} \cdot L \rightarrow \boxed{\vec{\theta}_C = -11,25 \cdot 10^{-3} \text{ rad}}$$



Reacciones: $\sum F_H = 0 \quad -H_A + \frac{P}{2} + \frac{P}{\sqrt{2}} = 0$

$$\boxed{H_A = 1,21P}$$

$$\sum F_V = 0 \quad R_A + R_C - \frac{P}{\sqrt{2}} = 0$$

$$\boxed{R_A = 0,143P}$$

$$\sum M^A = 0 \quad M_A + 2L R_C - \frac{P}{2} \cdot L - P \cdot \frac{L}{2} \sqrt{2} = 0$$

$$\boxed{M_A = 0,08 P \cdot L}$$

